AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2019

Biology Paper II

Time: 2 hours 10 minutes Marks: 50

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

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2. There are ELEVEN questions. Answer ALL questions. Questions 10 and 11 offer TWO choices. Attempt any ONE choice from each.

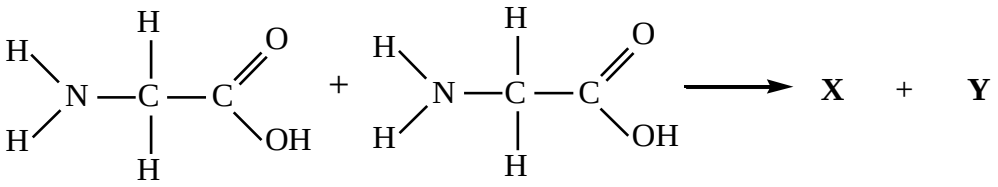
3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

a. Two amino acid molecules are shown in the given diagram. (2 Marks)



Illustrate the products, **X** and **Y**, from the given amino acid molecules.

Space for diagram

b. The given diagram represents a sequence of amino acids present in a polypeptide chain.

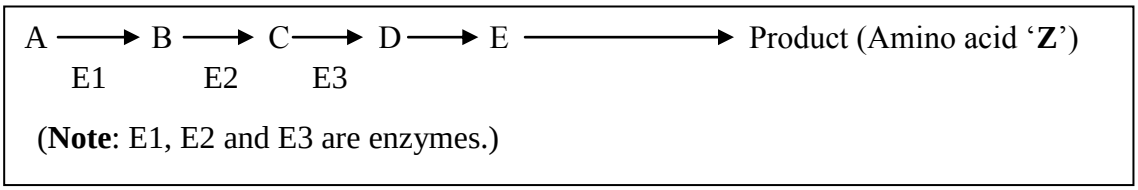


Will the function of the given polypeptide chain alter, if the amino acid B is replaced with an amino acid H? Elaborate your answer by using the example of sickle cell anaemia. (3 Marks)

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Q.2. (Total 2 Marks)

Given is a biochemical pathway involved in the synthesis of an amino acid '**Z**' in a cell.



a. Name the mechanism which regulates the amount of amino acid '**Z**' synthesised in the cell. (1 Mark)

_____

_____

b. How does the mechanism named in part (a) regulate the given biochemical pathway? (1 Mark)

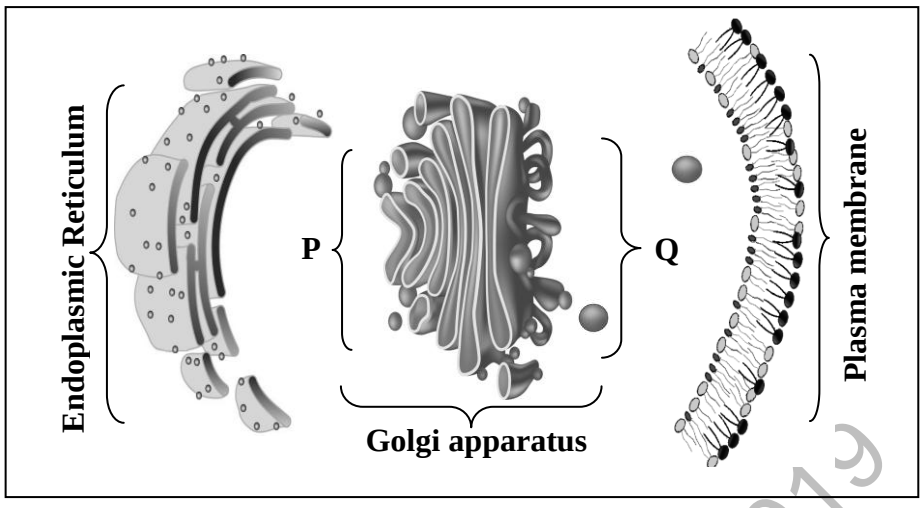
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Q.3. (Total 3 Marks)

The given diagram shows three parts of an animal cell.



a. Name the labelled side **P** of the Golgi apparatus. (1 Mark)

_____

_____

b. Why do **P** and **Q** face endoplasmic reticulum and plasma membrane respectively? (2 Marks)

_____

_____

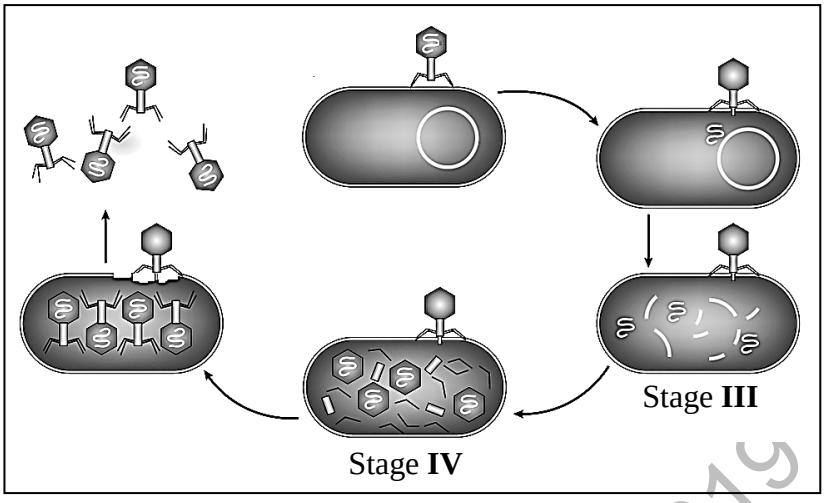
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Q.4. (Total 4 Marks)

The given diagram shows the life cycle of a bacteriophage.



a. Identify the type of life cycle shown in the given diagram. (1 Mark)

b. Give ONE reason to support your identification in part (a). (1 Mark)

c. Describe stage **III** and **IV** of the given life cycle. (2 Marks)

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Q.5. (Total 4 Marks)

a. Plants have evolved from green algae.

Mention any TWO evidences of this evolution. (2 Marks)

_____

_____

_____

_____

b. Describe the mode of reproduction of phylum deuteromycota of fungi. (2 Marks)

_____

_____

_____

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Q.6. (Total 5 Marks)

a. State any TWO characteristic features of bryophytes. (2 Marks)

b. Differentiate between Solanacea (potato family) and Fabaceae (pea family) with reference to their floral characteristics in the given table. (3 Marks)

Floral Characteristic	Solanacea	Fabaceae
Inflorescence		
Placentation		
Flower Symmetry		

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Q.7. (Total 4 Marks)

How does electron transport chain facilitate the production of adenosine tri phosphate (ATP) through chemiosmosis?

Q.8. (Total 5 Marks)

a. Name the secretory cells which are common to both, small intestine and large intestine. (1 Mark)

_____

_____

b. Differentiate between the large intestine and small intestine of human beings on the basis of their structure. (4 Marks)

S. No.	Small Intestine	Large Intestine
i.		
ii.		
iii.		
iv.		

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Q.9. (Total 3 Marks)

Complete the given table for tuberculosis.

Cause (Name of Bacteria)	Symptom
	i.
	ii.

Q.10 (Total 7 Marks)

EITHER

a.

i. Describe the structure attained by phospholipid molecules when they are placed in water. (2 Marks)

ii. Describe any FIVE functions of the phospholipid membrane (plasma membrane). (5 Marks)

OR

b. Reptiles have evolved to replace amphibians as dominant terrestrial animals.

i. Describe any FIVE adaptive features of reptile to the terrestrial environment. (5 Marks)

ii. Describe any TWO features of amphibians which limit their chances of survival in the terrestrial environment. (2 Marks)

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Q.11. (Total 8 Marks)

EITHER

a. The given tissues are present in the respiratory system of human beings.

- Elastic fibres
- Mucous glands
- Smooth muscles
- Ciliated epithelium

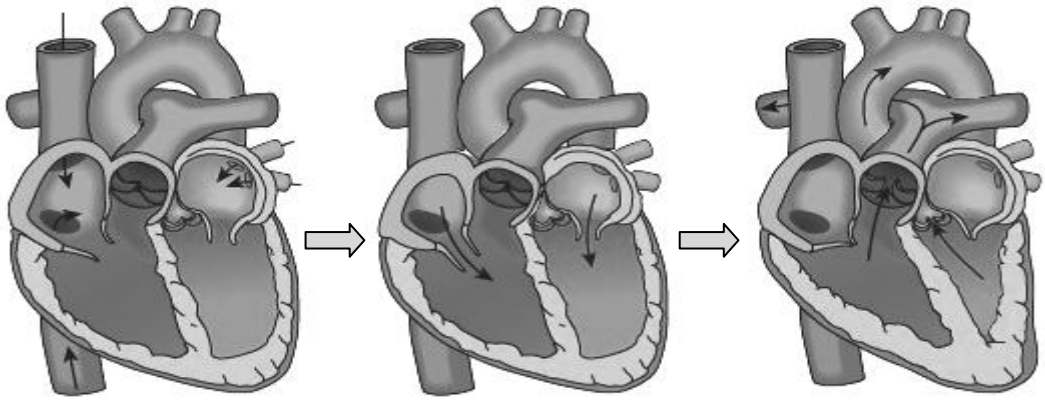
With reference to the respiratory system of human beings,

i. Name TWO organs which contain ALL of the given tissues. (2 Marks)

ii. Describe the function of any THREE of the given tissues. (6 Marks)

OR

b. The given diagrams represent phases of the cardiac cycle of human beings.



Phase I **Phase II** **Phase III**

i. Describe the given phases with reference to the blood pressure changes that occur in heart chambers. (6 Marks)

ii. Describe the regulation of cardiac cycle. (2 Marks)

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